

30 D Initial true count rate = $90 - 10 = 80$

No of half lives = $12/24 = 0.5$

$$C = 80(1/2)^{0.5} = 57 \text{ min}^{-1}$$

$$\text{Final count} = 57 + 10 = 67 \text{ min}^{-1}$$

